Esteban Marcelo Vizcaíno Gudiño

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Machine Learning Engineer

Telecommunications Engineer and Machine Learning Engineer (+450h intensive training). Experienced in designing, training, and deploying ML models (PyTorch, TensorFlow, XGBoost, Scikit-learn) and building robust backend services and data workflows (Python, FastAPI, PostgreSQL, Docker, Linux). Advanced practitioner in AI coding agents and autonomous developer tooling, accelerating high-quality software delivery, automated testing, and scalable architecture design. Available across time zones for remote roles.

Work Experience

Warehouse Manager | MANSUERA S.A. *May 2026 – Present*
Ibarra, Ecuador

- Built a Python/Flask warehouse management system from scratch and use it daily to organize and query approximately 3,000 inventory items sourced from MANSUERA operational data.
- Reduced a recurring inventory organization process from roughly **4–5 hours to 1–2 hours** using automated data imports, searchable records, and visual shelf maps; identified and removed invalid locations.
- Designed the PostgreSQL schema and deployed the database on Rocky Linux in a virtual machine, including user access, backups, network configuration, services, and firewall rules.
- Integrated a local Ollama assistant for inventory queries and Selenium automation for warehouse workflows.

Projects

Customer Churn Prediction API | Individual project *2026*

- Trained TensorFlow/Keras neural network and XGBoost classifiers to predict customer churn from telecom service data; compared both models on a **1,409-record test set**.
- Evaluated accuracy, precision, recall, F1, and ROC-AUC. XGBoost achieved **83.1% recall and 0.861 ROC-AUC**; exposed both models through a FastAPI prediction endpoint.

Adaptive AI Voice Interview Platform | Anyone AI final team project *2026*

- Designed interviewer prompts and a 1–5 answer-evaluation rubric; contributed to logic that selects questions from a stored bank or generates follow-up questions based on candidate responses through the OpenAI API.
- Worked remotely with four teammates using Git commits and pull requests. The application uses FastAPI, PostgreSQL, Redis-backed evaluation jobs, and Docker.

Adaptive Security for IoT | Undergraduate thesis, individual project *2025 – 2026*

- Developed an MQTT-based Edge AI system in C/C++ to select lightweight cryptographic algorithms from device telemetry; implemented a neural network, fuzzy-logic model, ESP32 firmware, and Orange Pi master node.
- Measured **54.57 μs** neural-network inference and **19.41 μs** fuzzy-logic inference. Thesis published by Universidad Técnica del Norte.

Skills

- **Machine Learning and Data:** PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, pandas, NumPy, model evaluation, SQL.
- **Programming and Infrastructure:** Python, C++, C, FastAPI, Flask, PostgreSQL, Redis, Docker, Git, Linux, Bash.
- **Applied AI and Agentic Tooling:** AI Coding Agents & Autonomous Workflows, OpenAI API, Ollama, prompt design, MQTT, ESP32, edge inference.
- **Languages:** Spanish (native), English (B1 CEFR; certified in 2024).

Education

Telecommunications Engineering Degree | Universidad Técnica del Norte *2021 – 2026*
Official Ecuadorian degree: Ingeniero en Telecomunicaciones; registered in the national higher-education system.

Machine Learning Engineer Career | Anyone AI *2026*
Completed 450+ hours of applied training and a remotely delivered final team project.

Certifications and Presentations

- **CCNA: Introduction to Networks** | Cisco, 2025.
- **Speaker, CIDU 2026** | Adaptive security for IoT, January 2026.
- **Speaker, AENIT 2026** | Dynamic selection of lightweight cryptographic algorithms for IoT, May 2026.